

## SUMMARY TABLE: BIOLOGY GRADE 11

### Sub-Strand 3.3: Excretion and Homeostasis in Animals — Teacher Reference (Pre-filled)

| SUMMARY TABLE: BIOLOGY GRADE 11 |                                                                                                                                                                      |
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| Sub-Strand                      | Sub-Strand 3.3: Excretion and Homeostasis in Animals                                                                                                                 |
| Driving Question                | DRIVING QUESTION: How do different animals maintain a balanced internal environment and eliminate toxic wastes despite drastic changes in their external conditions? |

### INSTRUCTIONS

FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

| Lesson # and Title                                                                                             | What did I observe?                                                                                                                                                                                                                                                                                                                                                          | What did I learn?                                                                                                                                                                                                                                                                                                                                                                    | How does this explain the phenomenon?                                                                                                                                                                                                                                                                                                                                                                                                      |
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| <b>Lesson 1</b><br><b>Same Hot Afternoon, Different Bodies: Launching the Excretion and Homeostasis Puzzle</b> | We observed that freshwater fish, marine fish, insects and birds have different excretory structures (gills and kidneys, Malpighian tubules, kidneys) and make different nitrogenous wastes (ammonia, urea, uric acid), and that a dog pants, a lizard sits still on a hot rock, goats produce concentrated urine and freshwater fish produce large amounts of dilute urine. | We learned that the type of nitrogenous waste an animal makes is linked to how much water is available to it: animals with plenty of water (freshwater fish) excrete toxic ammonia in dilute urine, while animals that must save water (insects and birds) excrete less toxic uric acid as a semi-solid paste, and marine cartilaginous fish (such as sharks) and mammals make urea. | We can explain that animals living in different environments have excretory structures adapted to remove nitrogenous wastes while conserving or removing water, but we cannot yet explain how a goat makes concentrated urine, why a dog pants, why a lizard basks, or how mammals such as humans filter their internal liquid wastes. That leads us to ask: How do mammals (like humans) process and filter their internal liquid wastes? |
| <b>Lesson 2</b><br><b>Inside the Goat Kidney: Cortex, Medulla, Pelvis and Nephrons</b>                         | We observed a goat/cow kidney with a dark outer cortex, a paler striped medulla, a hollow renal pelvis leading to the ureter, and blood vessels (renal artery and vein) entering and leaving at the hilum. Granular dots in the cortex suggested tiny filtering units.                                                                                                       | The mammalian kidney is organised into regions with different roles: the cortex contains many nephrons where blood is filtered, the medulla contains parts of nephrons and collecting ducts that help concentrate urine, and the pelvis collects urine and passes it to the ureter. The nephron is the functional unit of the kidney.                                                | Because the kidney receives a large blood supply and contains millions of nephrons arranged in the cortex and medulla, it can filter blood, remove urea and regulate water. This helps explain how a goat can drink water yet produce concentrated urine, while we still need to find out how nephrons turn filtered blood into urine and why freshwater fish produce dilute urine.                                                        |
| <b>Lesson 3</b><br><b>Following the Fluid: How the Nephron Filters Blood and Makes Urine</b>                   | We watched animations of the nephron and traced how blood, filtrate and urine differ in glucose, protein, urea and water. We saw that blood cells and proteins stay in the                                                                                                                                                                                                   | Urine formation involves ultrafiltration in the glomerulus (small molecules forced into Bowman's capsule under high pressure), followed by selective reabsorption of useful                                                                                                                                                                                                          | A goat that loses water on a hot day releases more ADH, so its collecting ducts reabsorb more water and it produces small amounts of concentrated urine. A                                                                                                                                                                                                                                                                                 |

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|                                                                                                                                                  | blood, glucose is fully reabsorbed in the proximal tubule, and urine is concentrated by water reabsorption in the loop of Henle and collecting duct.                                                                                                                                                                                                                                                                                  | substances and secretion of extra wastes and ions in the nephron tubule. ADH controls how much water is reabsorbed, so urine can be concentrated or dilute.                                                                                                                                                                                                                                                                                                                                                                                                           | freshwater fish gains water by osmosis, so it produces large amounts of dilute urine. New question: What other organs in mammals assist in excretion besides the kidney?                                                                                                                                                                                                                                                                                                                                          |
| <b>Lesson 4</b><br><b>Liver and Skin: Hidden Waste Processors Behind the Hot Afternoon</b>                                                       | We observed photomicrographs and charts of the liver (hepatocytes, sinusoids, bile canaliculi) and skin (epidermis, dermis, sweat glands, capillaries), and heard about cirrhosis, hepatitis and fatty liver disease.                                                                                                                                                                                                                 | The liver deaminates excess amino acids to make urea, breaks down haemoglobin to make bile pigments and processes toxic substances; the skin excretes sweat containing water, salts and a little urea and helps cool the body. Liver disorders damage these functions.                                                                                                                                                                                                                                                                                                | Urea in the goat's urine begins in the liver, is carried by blood to the kidney and removed, while sweat or panting helps animals lose heat on a hot afternoon. The liver and skin work with the kidney to remove wastes and keep the body stable, but we still need to explain how water, temperature and sugar levels are kept constant.                                                                                                                                                                        |
| <b>Lesson 5</b><br><b>Feedback Loops: How the Body Keeps Water and Blood Glucose Steady</b>                                                      | We observed that hormone signals such as ADH, insulin and glucagon form feedback loops, and that in diabetes mellitus and diabetes insipidus the symptoms (thirst, lots of urine, sugar in urine or very dilute urine) show where a loop has failed.                                                                                                                                                                                  | We learned that negative feedback detects a change from a set point and triggers a response that reverses it; ADH adjusts how much water the kidney reabsorbs, and insulin and glucagon keep blood glucose within a narrow range.                                                                                                                                                                                                                                                                                                                                     | We can explain why the goat can produce concentrated urine to save water while the freshwater fish excretes dilute urine, how the body keeps water and glucose steady after exercise and a meal, and why the next question is how body temperature is kept stable in animals like the dog and the lizard.                                                                                                                                                                                                         |
| <b>Lesson 6</b><br><b>Keeping Cool and Staying Warm: Thermoregulation in the Dog, the Lizard and Us</b>                                          | We observed a dog panting and cattle and goats resting in shade at midday, an agama lizard basking on a rock in the morning and moving into shade or a crevice at midday, and, in video clips and notes, animals that burrow, hibernate or aestivate during extreme cold or dry conditions. We also saw diagrams of blood vessels in the skin widening (vasodilation) and narrowing (vasoconstriction) and of sweating and shivering. | Body temperature is kept within a narrow range by negative feedback: thermoreceptors detect a change, the hypothalamus acts as control centre, and effectors such as sweat glands, skin arterioles and muscles reverse the change. Endotherms like dogs, goats and humans use physiological responses (sweating, panting, vasodilation, vasoconstriction, shivering) plus behaviour, while ectotherms like lizards rely mainly on behaviour (basking, shade-seeking, burrowing). Hibernation and aestivation help animals survive prolonged cold or hot, dry periods. | The dog pants because evaporation of water from its tongue and airways carries heat away, and its skin blood vessels dilate to lose heat, keeping its body temperature steady. The lizard stays still on the warm rock because it cannot control its temperature internally, so it gains heat by basking and moves into shade when it gets too hot. Both are using homeostasis to survive heat, which raises the question of why excretion and homeostasis are so vital for the survival of all living organisms. |
| <b>Lesson 7</b><br><b>Putting It All Together: Final Explanation of How Animals Keep a Balanced Internal Environment and Remove Toxic Wastes</b> | What we observed: the panting dog, the basking lizard, goats producing concentrated urine and freshwater fish producing large volumes of dilute urine; the layers of the goat or cow kidney (cortex, medulla, pelvis); wastes in insects, birds, fish and mammals; and how feedback loops work.                                                                                                                                       | What we learned: homeostasis keeps temperature, water, salts and blood sugar steady using negative feedback; excretion removes toxic metabolic wastes such as ammonia, urea and uric acid; the kidney, liver and skin do this work in mammals; and behaviours such as basking, panting, burrowing, hibernation and aestivation help animals cope with environmental change.                                                                                                                                                                                           | Our final explanation: Animals face different environments, but all must keep their cells in a stable internal environment and remove toxic wastes. The dog pants to lose heat; the lizard basks or shelters to adjust its temperature behaviourally; the goat's nephrons reabsorb water to make concentrated urine; the freshwater fish gains water by osmosis and excretes dilute                                                                                                                               |

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|  |  |  | urine. Because each structure or behaviour suits the animal's habitat, its cells survive and it stays healthy. This is why excretion and homeostasis are essential. |
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ATTRIBUTION AND LICENSING

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